1. One-Shot Prompting

One-shot prompting is a **prompting technique in generative AI** where you instruct a model to perform a task **by providing exactly one example .**

Formula

Task Instruction + One-Example + Input

The example helps the model understand:

- The task
- The expected reasoning
- The response style

Why It Is Used

Sometimes a task is ambiguous.

For example:

Classify sentiment.

The model may understand the task, but may not understand:

- Output format
- Label names
- Expected style

By providing **one example**, we guide the model.

Example

Without Example (Zero-Shot)

Classify sentiment.

Review: This phone is awesome.

Output:

Positive

or

The sentiment is positive.

or

This review expresses a positive opinion.

The format may vary.

With One Example (One-Shot)

Classify sentiment.

Example:
Review: The product is terrible.
Sentiment: Negative

Now classify:

Review: This phone is awesome.

Output:

Sentiment: Positive

The example teaches the model the desired format.

How It Works

Suppose we have:

Translate English to French.

Example:
English: Good Morning
French: Bonjour

Now translate:

English: Thank You

Step 1: Model Reads Instruction

Task = Translation

Step 2: Model Reads Example

English: Good Morning
French: Bonjour

The model learns:

Input Pattern:
English: X

Output Pattern:
French: Y

Step 3: Model Applies Pattern

Input:

English: Thank You

Output:

French: Merci

Step 4: Response Generated

French: Merci

Advantages

1. Better Accuracy

One example gives additional guidance.

2. More Consistent Output

The model follows the demonstrated format.

3. Better Structured Responses

Useful for:

- JSON generation
- SQL generation
- Report generation

Limitations

1. More Tokens

Prompt becomes longer.

2. Example Bias

The model may overfit to the provided example.

3. Not Enough for Complex Tasks

Some tasks require multiple examples (Few-Shot).

Langchain_groq example:

```
from langchain_groq import ChatGroq
from langchain.prompts import PromptTemplate

llm = ChatGroq(
    api_key="gsk_xxxxxxxxxxxxx",
    model="llama-3.3-70b-versatile",
    temperature=0
)

template = """
Classify sentiment.

Example:
Review: The food was awful.
Sentiment: Negative

Now classify:

Review: {review}
"""

prompt = PromptTemplate(
    input_variables=["review"],
    template=template
)
```

```
chain = prompt | llm
 result = chain.invoke({
   "review": "The laptop performance is excellent."
})
print(result.content)
```

Real Industry Use Cases

Resume Skill Extraction

Example Resume → Example JSON
New Resume → Extract JSON

SQL Generation

Example Question → Example SQL
New Question → Generate SQL

Few-Shot Prompting

Few-shot prompting is a **prompting technique in generative AI** where you instruct a model to perform a task **by providing multiple examples (typically 2–10)** before asking it to perform the actual task.

Instead of learning from just one example (One-Shot), the model learns the pattern from several examples and then applies it to a new input.

Formula

Instruction

+

Example 1

+

Example 2

+

Example 3

...

+

New Input

Why It Is Used

Some tasks are too complex for Zero-Shot or One-Shot.

Multiple examples help the model understand:

- The task
- The format
- The reasoning pattern
- Edge cases
- Special rules

As the number of examples increases, the model usually becomes more accurate.

Example 1: Sentiment Analysis

Prompt

Classify sentiment.

Example 1:
Review: This phone is amazing.
Sentiment: Positive

Example 2:
Review: Worst laptop ever.
Sentiment: Negative

Example 3:
Review: The package arrived today.
Sentiment: Neutral

Now classify:

Review: The battery life is fantastic.

Output

Sentiment: Positive

The model learned the pattern from three examples.

How It Works

Suppose we want the model to identify whether a number is Even or Odd.

Prompt

Example 1:
2 → Even

Example 2:
7 → Odd

Example 3:
10 → Even

Now:
15 →

Step 1: Read Examples

The model observes:

2 → Even
7 → Odd
10 → Even

Step 2: Detect Pattern

It infers:

Divisible by 2 → Even
Otherwise → Odd

Step 3: Apply Pattern

15

is not divisible by 2.

Step 4: Generate Output

Odd

Why Few-Shot Often Works Better

Imagine teaching a child.

Zero-Shot

Solve this type of problem.

No example.

One-Shot

Here is one solved example.
Now solve this.

Better.

Few-Shot

Here are 5 solved examples.
Now solve this.

Much better.

The same idea applies to LLMs.

Advantages

1. Higher Accuracy

More examples = clearer pattern.

2. Reduced Hallucinations

Examples constrain the model.

3. Handles Complex Tasks

Works better for:

- Classification
- Extraction
- Data transformation
- Business reports

Limitations

1. Higher Token Cost

Prompt becomes larger.

2. Slower Responses

More tokens to process.

3. Context Window Usage

Too many examples can consume the context window.

Better LangChain Version Using FewShotPromptTemplate

```
from langchain_groq import ChatGroq
from langchain.prompts import (
    FewShotPromptTemplate,
    PromptTemplate
)

llm = ChatGroq(
    api_key="gsk_xxxxxxxxxxxxx",
    model="llama-3.3-70b-versatile",
```

```
  temperature=0
)

examples = [
  {
    "review": "This phone is amazing.",
    "sentiment": "Positive"
  },
  {
    "review": "Worst laptop ever.",
    "sentiment": "Negative"
  },
  {
    "review": "The package arrived today.",
    "sentiment": "Neutral"
  }
]

example_prompt = PromptTemplate(
  input_variables=["review", "sentiment"],
  template="""
Review: {review}
Sentiment: {sentiment}
"""
)

few_shot_prompt = FewShotPromptTemplate(
  examples=examples,
  example_prompt=example_prompt,
  prefix="Classify sentiment.",
  suffix="Review: {review}",
  input_variables=["review"]
)
```

```
chain = few_shot_prompt | llm

result = chain.invoke({
    "review": "The camera quality is fantastic."
})

print(result.content)
```